PHYS 7501:
Particles and Nuclei I

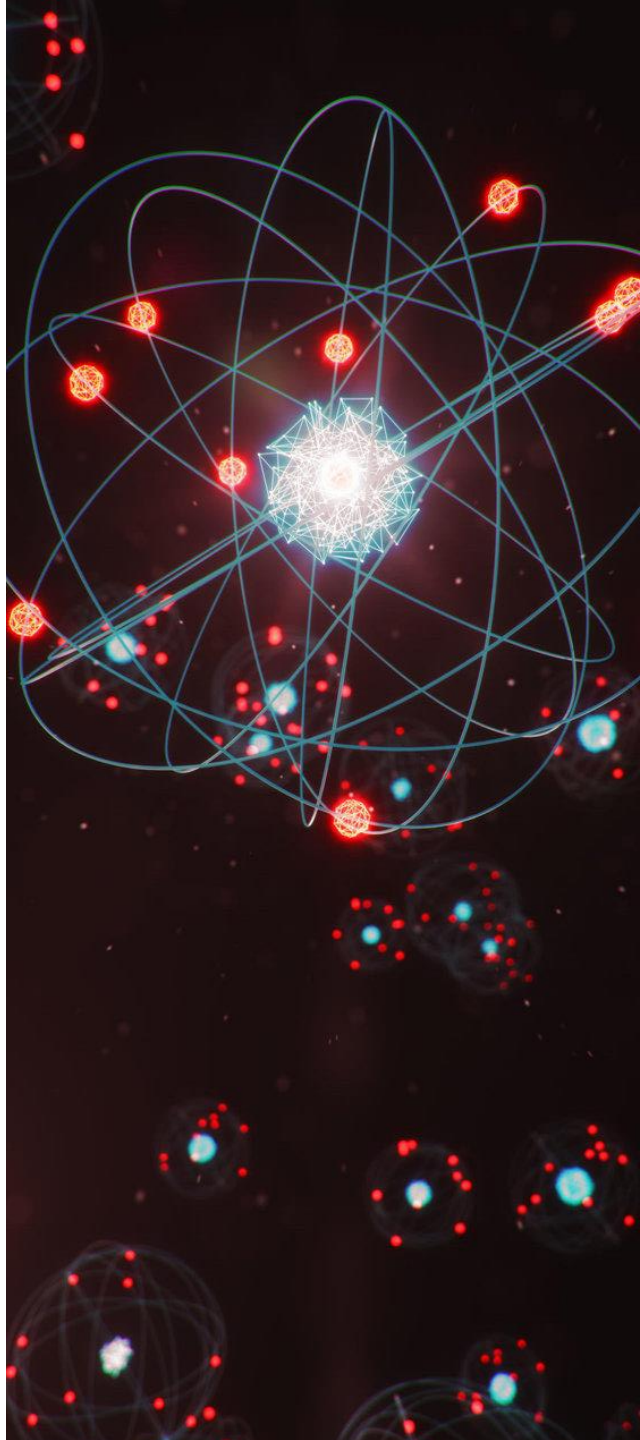
Tues./Thurs. 2 – 3:20PM
Clippinger 105

Dr. Andrea Richard

Edwards Accelerator Lab 204

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Available via e-mail & Teams

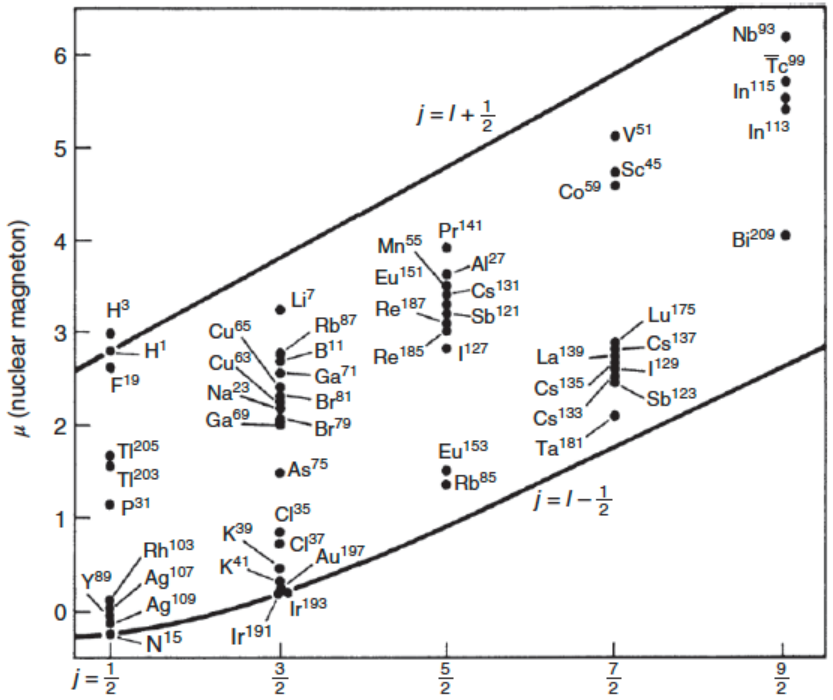


HW#2 Problem #5

- Schmidt Limits Equation in the Textbook are incorrect....
- They should be:

For j $= \ell + s$**:** $\mu = g_\ell \left(j - \frac{1}{2}\right) + \frac{1}{2} g_s$

For j $= \ell - s$**:** $\mu = \left(\frac{j(j+\frac{3}{2})}{j+1}\right) g_\ell - \left(\frac{j}{2(j+1)}\right) g_s$



Your result should look like this ^

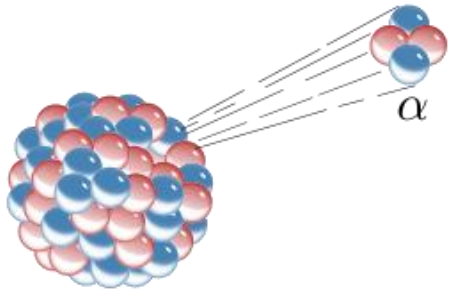
ALPHA DECAY

Modern Nuclear Chemistry, Chapter 7

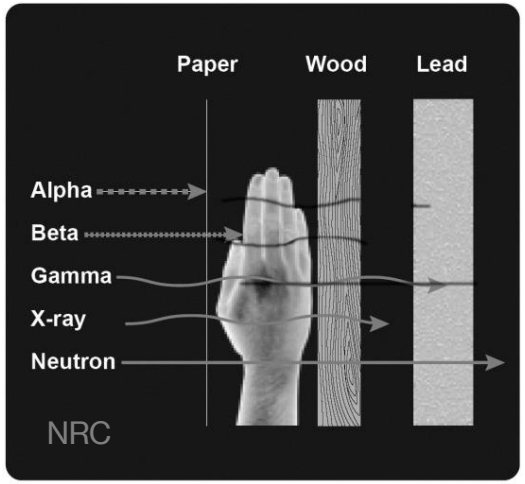
Monday	Tuesday	Wednesday	Thursday	Friday
9/22	9/23	9/24	9/25	9/26
	Alpha Decay		**Beta Decay** HW #2 Due	
9/29	9/30	10/1	10/2	10/3
	Gamma Decay		**Fission**	
10/6	10/7	10/8	10/9	10/10
	Exam Day		**Nuclear Forces** HW#3 Due	

The Historical Perspective

- Ernest Rutherford and collaborators examined samples of uranium and radium, finding that they emitted energetic particles



- In 1899, Rutherford studied the penetrating power of radiation from uranium and found that some energetic particles were stopped after a thin material and some took much more material to be stopped
  - He naturally ranked them as α, β in order of their penetrating power



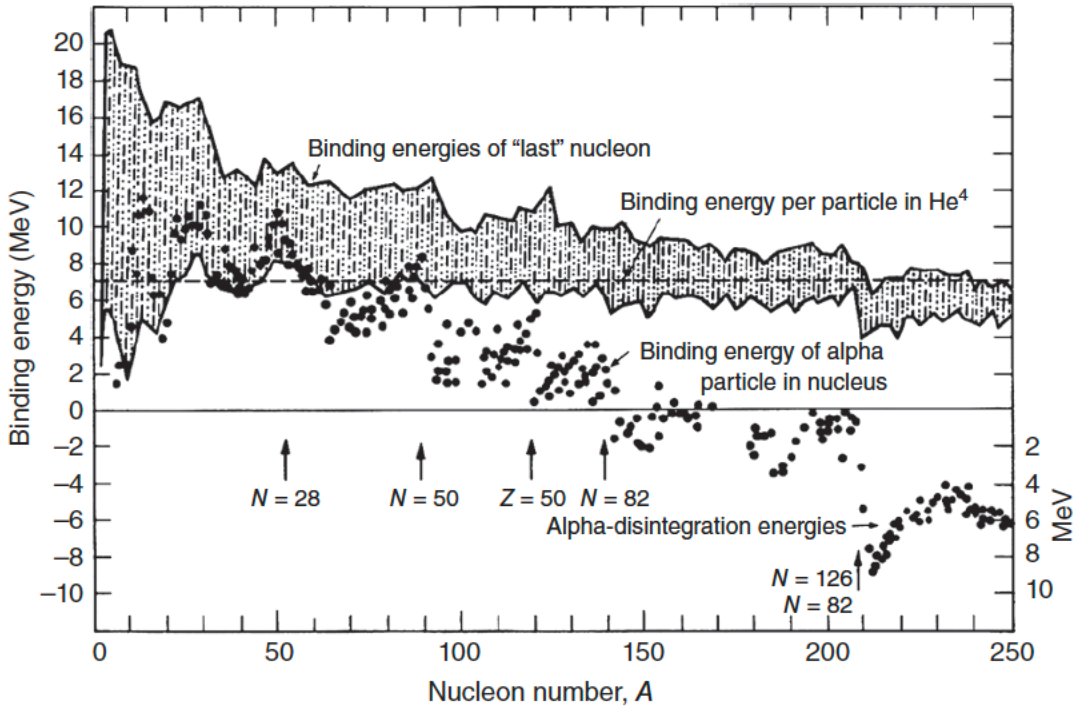
- In 1909, he showed that these so-called "α-rays" were, in fact, ${}^{4}_{2}He_{2}$ nuclei.

- Many heavy nuclei decay via α-emission, but not other nucleons (deuteron, triton, etc.)
  - Why is this the case? What is so special about α decay?

α-Decay Basics

- Alpha decay is a spontaneous process by which heavy nuclei emit an $\boldsymbol{\alpha}$ particle moving the system to a lower energy state
- The process of alpha-decay can be written as: $\mathbf{{}^{A}_{Z}Z_{N} \rightarrow {}^{A-4}_{Z-2}Z_{N-2} + {}^{4}_{2}He_{2} + Q_{\alpha}}$
  - Example: ${}^{238}\mathrm{U} \rightarrow {}^{234}\mathrm{Th} + {}^{4}\mathrm{He} + Q_{\alpha}$
- The value of $\boldsymbol{Q_{\alpha}}$ (Q-value) is positive, or endothermic, for spontaneous $\boldsymbol{\alpha}$ decay

$$\boldsymbol{Q_{\alpha} = \Delta_{parent} - \Delta_{daughter} - \Delta_{^4He}}$$

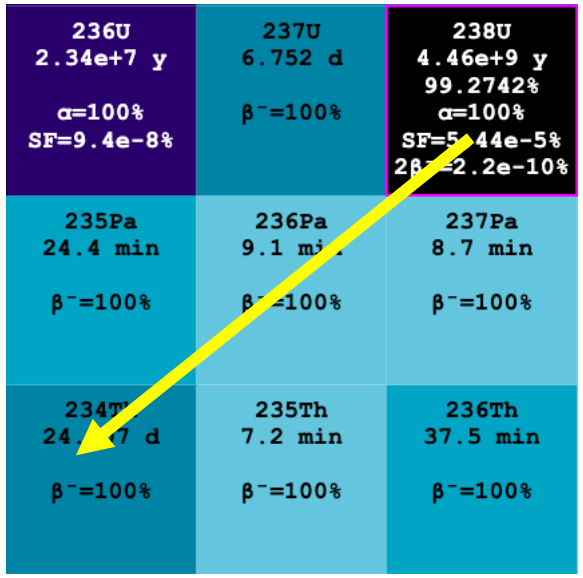


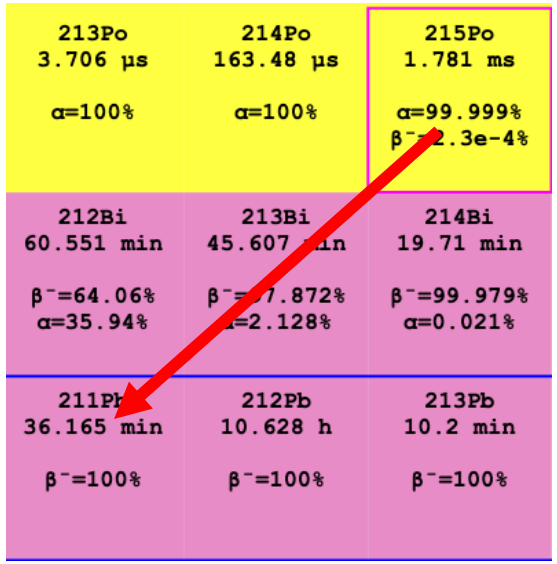
-Why does it make sense for $\boldsymbol{Q_{\alpha}}$ to be positive?

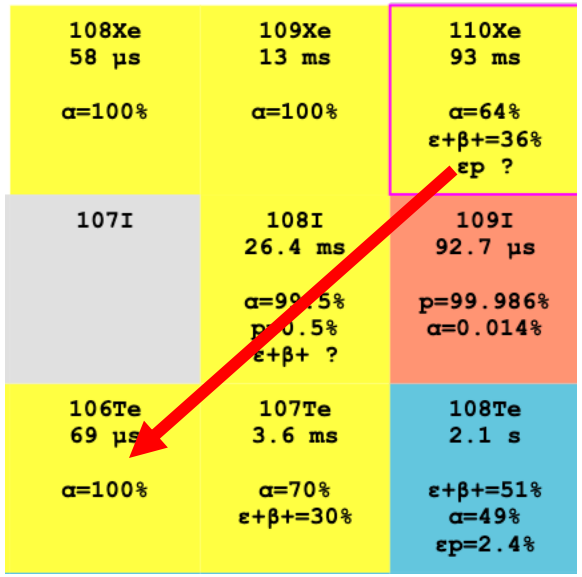
-In terms of the SEMF, losing 2 protons lowers the Coulomb energy, doesn't impact asymmetry or pairing, and minimally changes the surface and volume terms (per A)

-While the last nucleon is tightly bound, so is a nucleon a ^{4}He nucleus

-It is not nearly as bound as the nucleus itself, so $\boldsymbol{\alpha}$ emission is energetically favorable above A~150

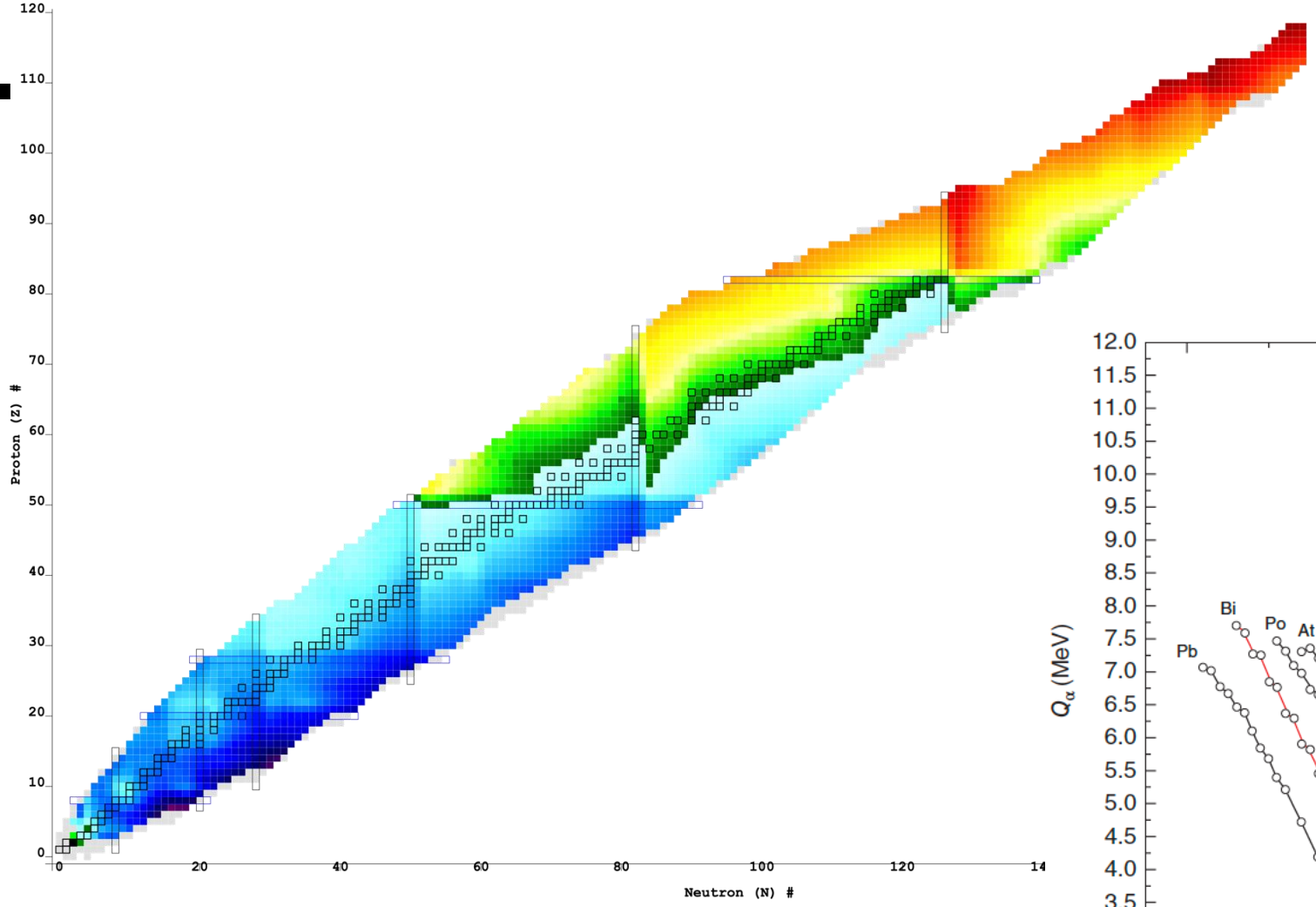


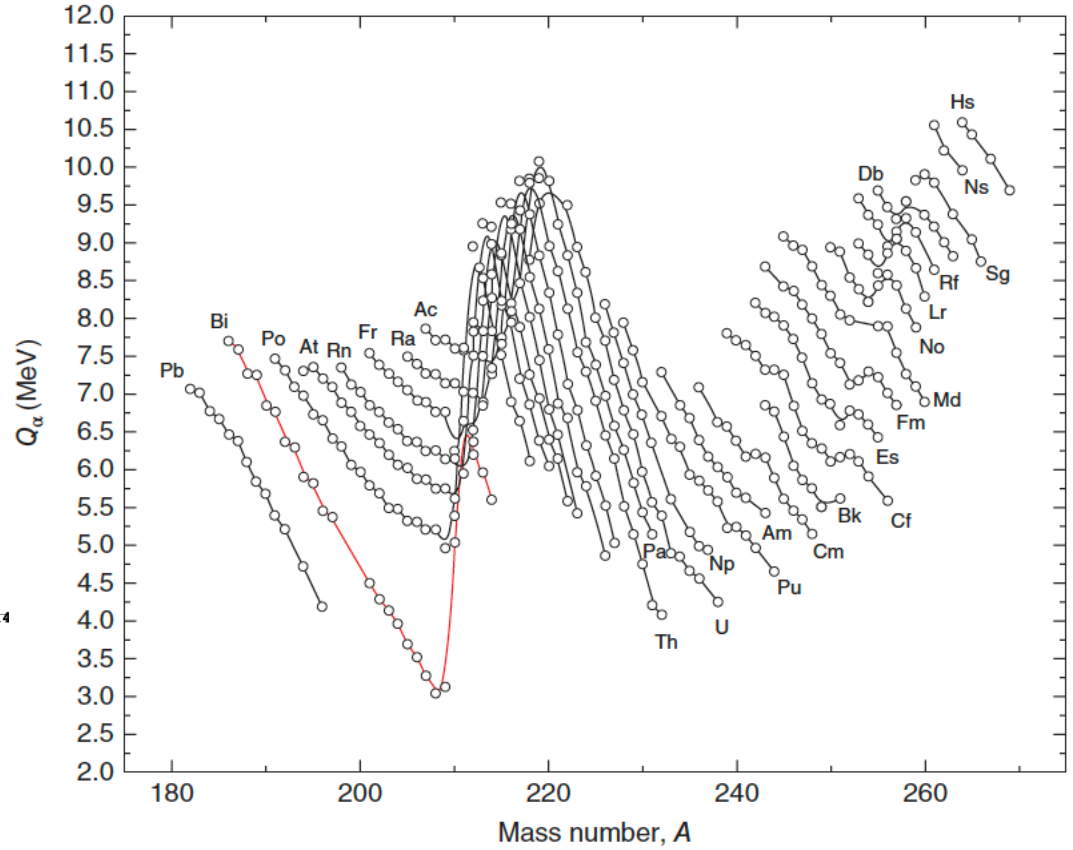




- Take a few minutes to investigate if the following nuclei will undergo α decay, if so, calculate the Q-value
  - ^{182}Hf **Yes, but No!**
  - ^{60}Fe **No!**
  - ^{110}Xe **Yes!** $Q_\alpha = \Delta_{110Xe} - \Delta_{106Te} - \Delta_{4He} = 3872\ \text{keV}$
  - ^{215}Po **Yes!** $Q_\alpha = \Delta_{215Po} - \Delta_{211Pb} - \Delta_{4He} = 7526.3\ \text{keV}$

Magic Number Trends





Energy Emitted from α Decay

- When an α is emitted, it will share energy with the heavy recoil – what is the relationship between kinetic energy and Q_α?
- Conservation of energy and momentum:

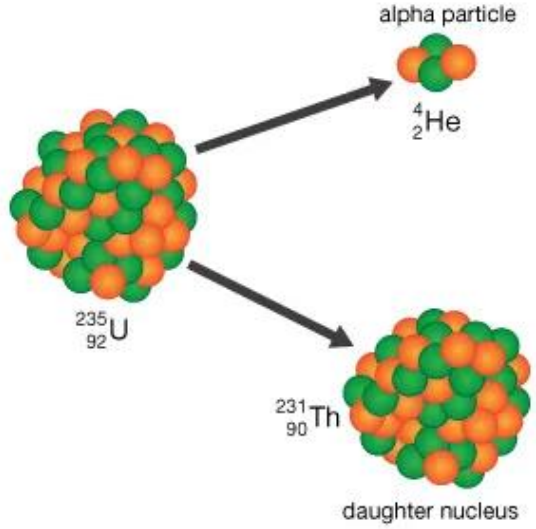
$$\vec{\mathbf{p}}_{\mathbf{parent}} = \vec{\mathbf{p}}_{\mathbf{daughter}} + \vec{\mathbf{p}}_{\boldsymbol{\alpha}}$$

- But $\vec{\mathrm{p}}_{\mathrm{parent}} = 0$ because the parent is at rest. So, the daughter and α will move in opposite directions: $\vec{\mathrm{p}}_{\mathrm{daughter}} = -\vec{\mathrm{p}}_{\alpha}$
- Then for energy:

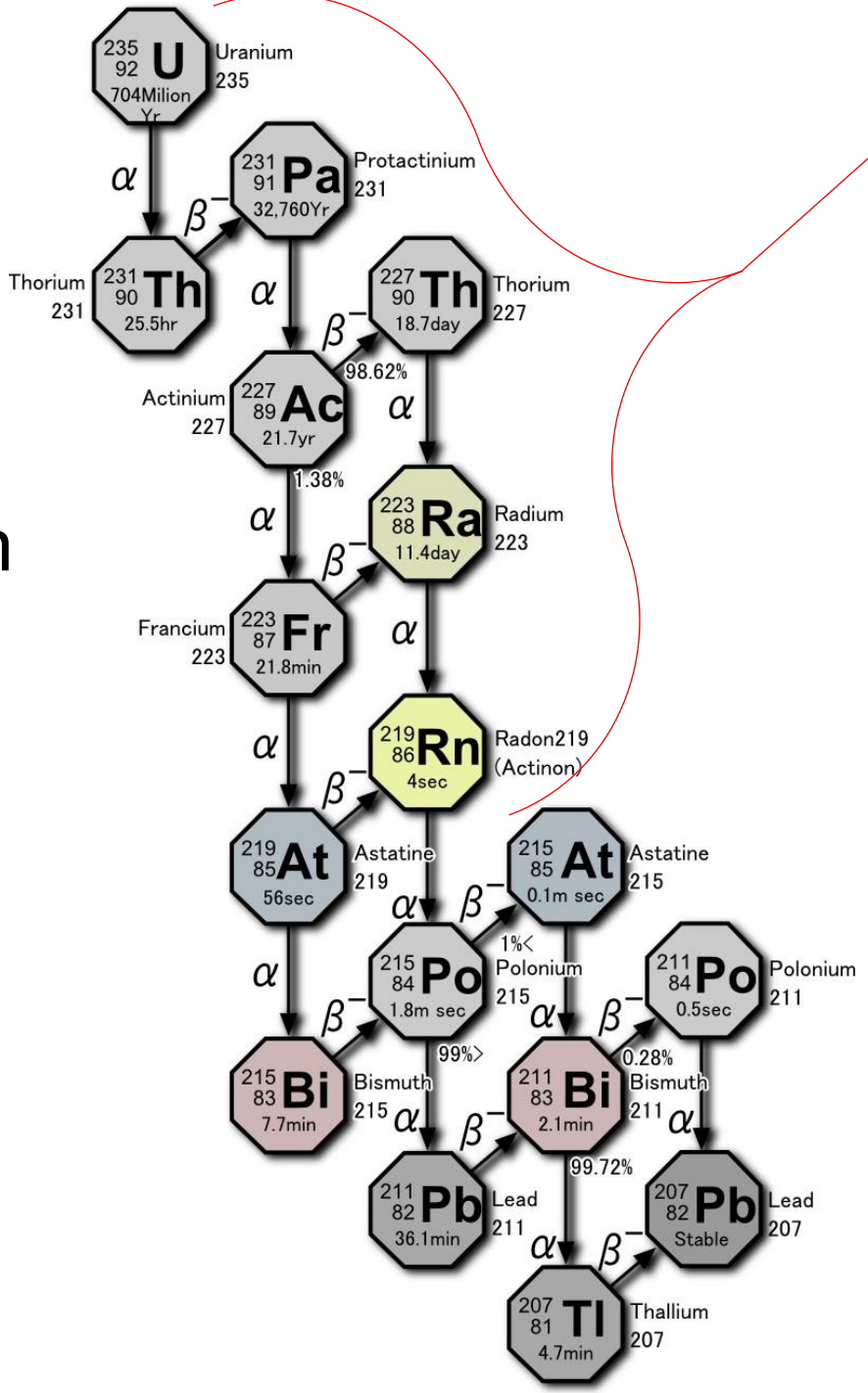
$$\frac{\boldsymbol{p_{parent}^2}}{\boldsymbol{2A_{parent}}} + \boldsymbol{Q_\alpha} = \frac{\boldsymbol{p_{daughter}^2}}{\boldsymbol{2A_{daughter}}} + \frac{\boldsymbol{p_\alpha^2}}{\boldsymbol{2A_\alpha}} = \frac{\boldsymbol{p_\alpha^2}}{\boldsymbol{2A_{daughter}}} + \frac{\boldsymbol{p_\alpha^2}}{\boldsymbol{2A_\alpha}} = \frac{\boldsymbol{A_\alpha}}{\boldsymbol{A_{daughter}}}\boldsymbol{KE_\alpha} + \boldsymbol{KE_\alpha}$$

- $Q_\alpha = \frac{A_\alpha + A_{daughter}}{A_{daughter}} KE_\alpha = \frac{A_{\mathrm{parent}}}{A_{daughter}} KE_\alpha$
- So, $\boldsymbol{KE_\alpha = \frac{A_{daughter}}{A_{parent}} Q_\alpha}$

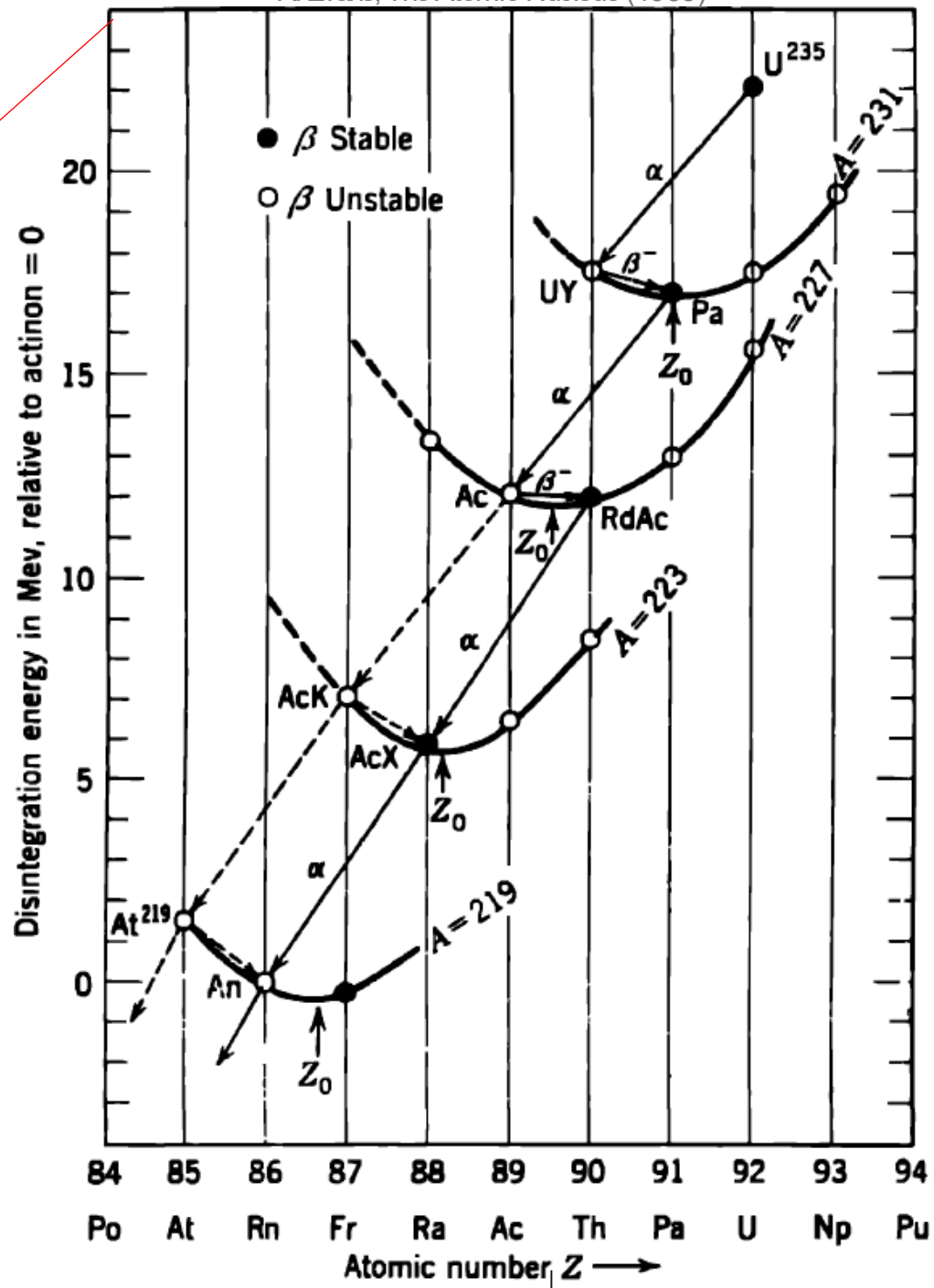
This is a small effect overall



Aside: Decay sequences can be mapped by considering the mass excess of nuclides involved, E.g. 235U

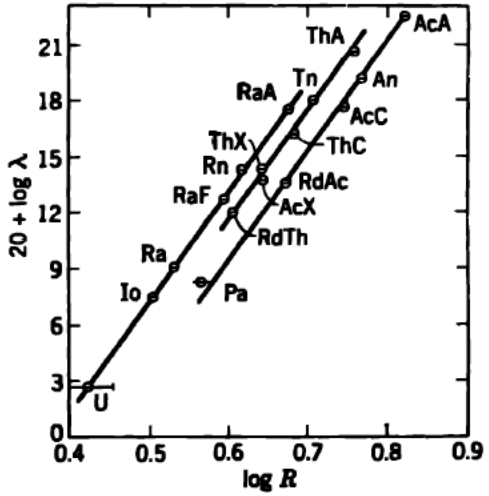


R. Evans, The Atomic Nucleus (1955)



Geiger-Nuttall Relation

Geiger & Nuttall, Philisoph. Mag. (1912)

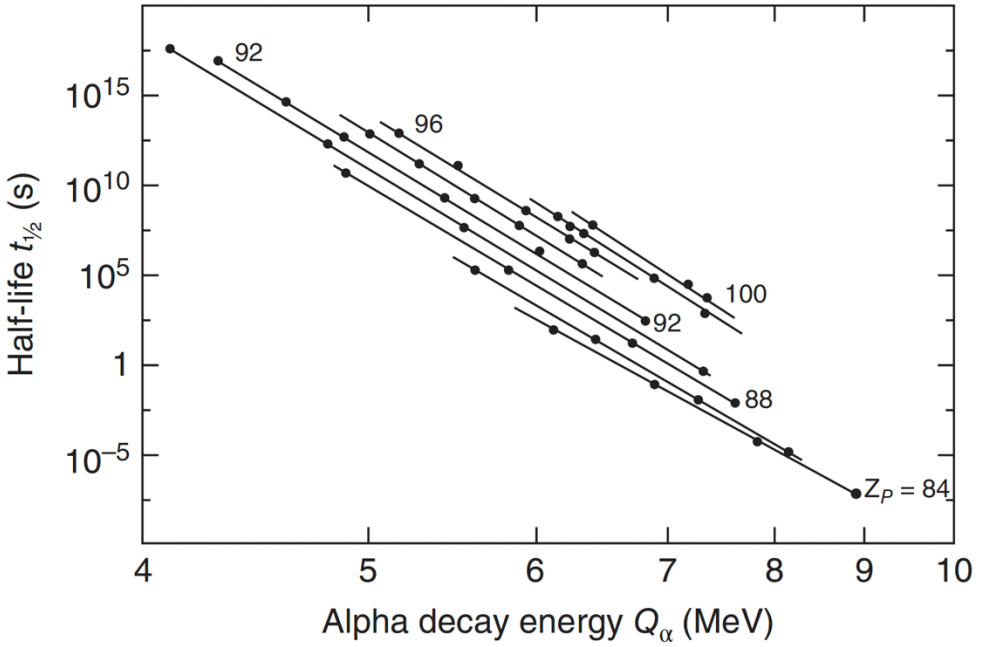


- In an early effort to characterize α decay, Geiger & Nuttall compared the range of α particles in a material vs. $t_{1/2}$ of the α emitter (source)
  - They discovered a linear relationship in log-log space
- Using Q_α instead of range, we get the Geiger-Nuttall relation:

$$log_{10}(t_{1/2}) = A + \frac{B}{\sqrt{Q_\alpha}}$$

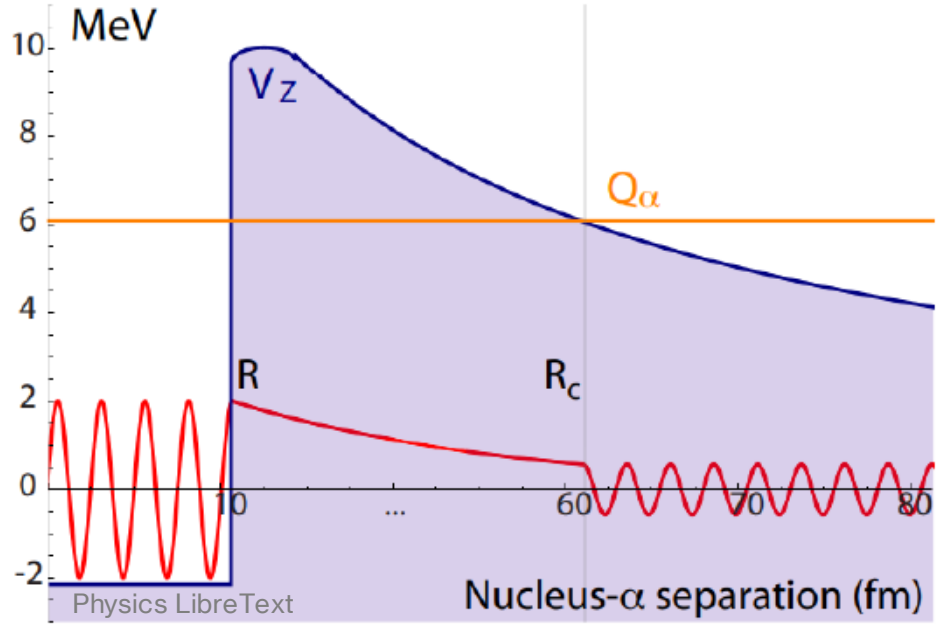
A and B have Z dependence

- α energy strongly impacts the $t_{1/2}$
- For a $\times 2$ increase in Q_α nearly 20 orders of magnitude decrease in $t_{1/2}$



Theory of α Emission

- Cannot be understood classically
- In 1928, almost simultaneously, Gamow and Gurney+Condon developed the theory to describe tunneling
- Inside the nucleus, the potential is attractive
- Outside the nucleus, the $\boldsymbol{\alpha}$ −particle feels repulsion from the Coulomb force

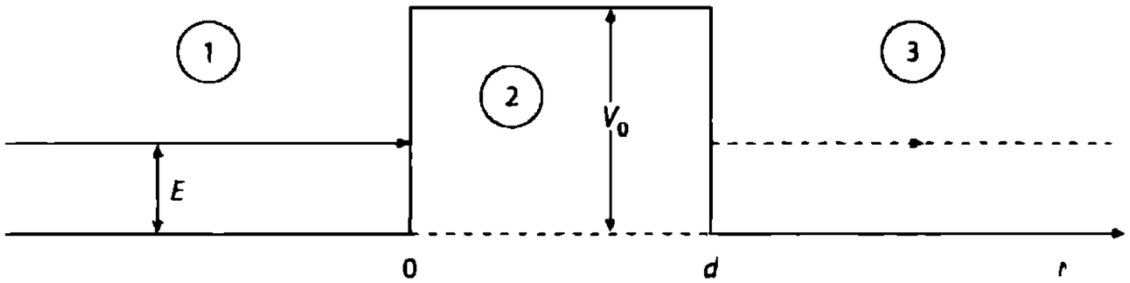


- When an $\boldsymbol{\alpha}$ −particle hits the Coulomb barrier, formed by the nuclear core (daughter from the decay), it has some probability for tunneling
- The challenge is to calculate the transmission probability (P_α)
  - $P_\alpha = \lambda = \mathrm{fT}$

λ = disintegration constant
f = frequency at which the particle hits the barrier
T = transmission coefficient (probability for crossing the barrier)

Tunneling through a 1D Square Barrier

- Consider the simplest tunneling case: plane wave penetrating a square barrier
- Particle with mass $\boldsymbol{m}$ and energy $\boldsymbol{E}$ hitting a barrier of width $\boldsymbol{a}$ (*d*) and height $\boldsymbol{V_0 > E}$ can be described by three regions:



1. Incidence and Reflection

$$\boldsymbol{\psi_1 = Ae^{ikx} + Be^{-ikx}}$$

2. Decaying in Barrier and Decaying Reflection

$$\boldsymbol{\psi_2 = Ce^{-\kappa x} + De^{\kappa x}}$$

3. Outgoing

$$\boldsymbol{\psi_3 = Fe^{ikx}}$$

Where the wavenumber $\mathbf{k}$ is from $\boldsymbol{\hbar^2 k^2 = 2mE}$ and the decay constant $\boldsymbol{\kappa}$ is from $\boldsymbol{\hbar^2 \kappa^2 = 2m(V_0 - E)}$

- The probability to make it through the barrier is the Transmission Coefficient: $\boldsymbol{T = \left|\frac{F}{A}\right|^2}$
- In Quantum Mechanics, we know that

$$\boldsymbol{T = \left|\frac{2\kappa k e^{-ika}}{2\kappa k\, cosh(\kappa a) - i(k^2 - \kappa^2)\, sinh(\kappa a)}\right|^2}$$

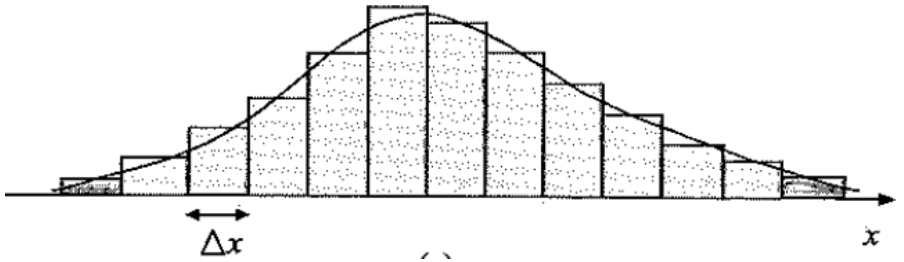
- For large $\boldsymbol{\kappa a}$, $\boldsymbol{sinh(\kappa a) \approx cosh(\kappa a) \approx \frac{1}{2} e^{\kappa a}}$
- So,

$$\boldsymbol{T = \left|\frac{4\kappa k}{(k^2 + \kappa^2)}\right|^2 e^{-2\kappa a}}$$

Exponent dominates so we write, $\mathrm{T} \approx \boldsymbol{e^{-2\kappa a} = e^{-2\mathrm{G}}}$

Tunneling through an arbitrary barrier

B.Martin, Nuclear and Particle Physics (2009)



- The result from the 1D barrier can be generalized by breaking the barrier into a series of 1D barriers
  - This is a trick known as the WKB approximation (Wentzel, Kramers, Brillouin)
- Replace $2\kappa a$ with $2\sum \kappa(x)\Delta x$ and use an integral sum instead of a Riemann sum

$$2\kappa a \rightarrow 2\int \sqrt{(\frac{2m}{\hbar^2}[V(x)-E])}\mathrm{dx}$$

- This is then generalized to 3D:

$$2G = \frac{2}{\hbar}\int \sqrt{(2m\,[Vc(r)-E])}dr$$

G is the Gamow factor

General details:

- m is the reduced mass: $\mu = \frac{m_\alpha m_{daughter}}{m_\alpha + m_{daughter}}$
- The limits for integration will be from the border of the potential well $R = r_0(A_\alpha^{1/3} + A_{daughter}^{1/3})$ to the classical distance of closest approach b, where $E = Q_\alpha = \frac{Z_\alpha Z_{daughter}}{b}e^2$, so $\mathbf{b} = \frac{Z_\alpha Z_{daughter}}{Q_\alpha}e^2$
- $V_c = \frac{Z_\alpha Z_{daughter}}{r}e^2$
- So, $2G = \frac{2}{\hbar}\sqrt{2\mu Q_\alpha}\int_b^R(\frac{b}{r})^{1/2}dr =$

$$2\frac{e^2}{\hbar c}Z_\alpha Z_{daughter}\sqrt{\frac{2\mu c^2}{Q_\alpha}}\left[\cos^{-1}(\sqrt{\frac{R}{b}}) - \sqrt{\frac{R}{b}(1-\frac{R}{b})}\right]$$

Why should you care? Because now you can estimate the transmission coefficient for any arbitrary case.

Tunneling through a thick barrier

- The transmission coefficient is $\mathrm{T} = e^{-2\mathrm{G}}$, where

This is pretty complicated

$$2G = \frac{2}{\hbar}\sqrt{2\mu Q_\alpha}\int_b^R \left(\frac{b}{r}\right)^{1/2} dr = 2\frac{e^2}{\hbar c}Z_\alpha Z_{daughter}\sqrt{\frac{2\mu c^2}{Q_\alpha}}\left[\cos^{-1}\left(\sqrt{\frac{R}{b}}\right) - \sqrt{\frac{R}{b}\left(1-\frac{R}{b}\right)}\right]$$

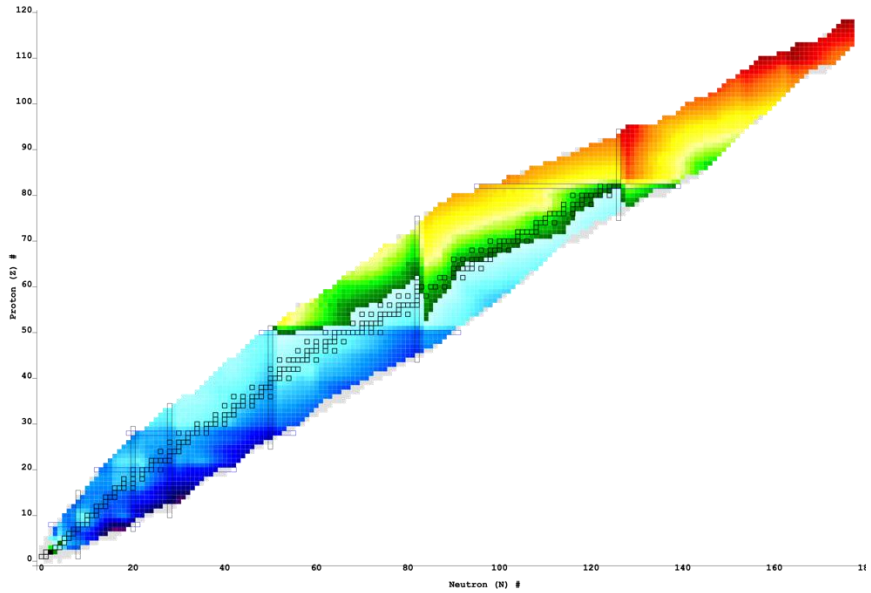
- Conveniently, most of the time $b \gg R$ so $\left[cos^{-1}\left(\sqrt{\frac{R}{b}}\right) - \sqrt{\frac{R}{b}\left(1-\frac{R}{b}\right)}\right]$ reduces to $\frac{\pi}{2}\left[1-\frac{4}{\pi}\sqrt{\frac{R}{b}}\right]$
- So, the Gamow factor becomes

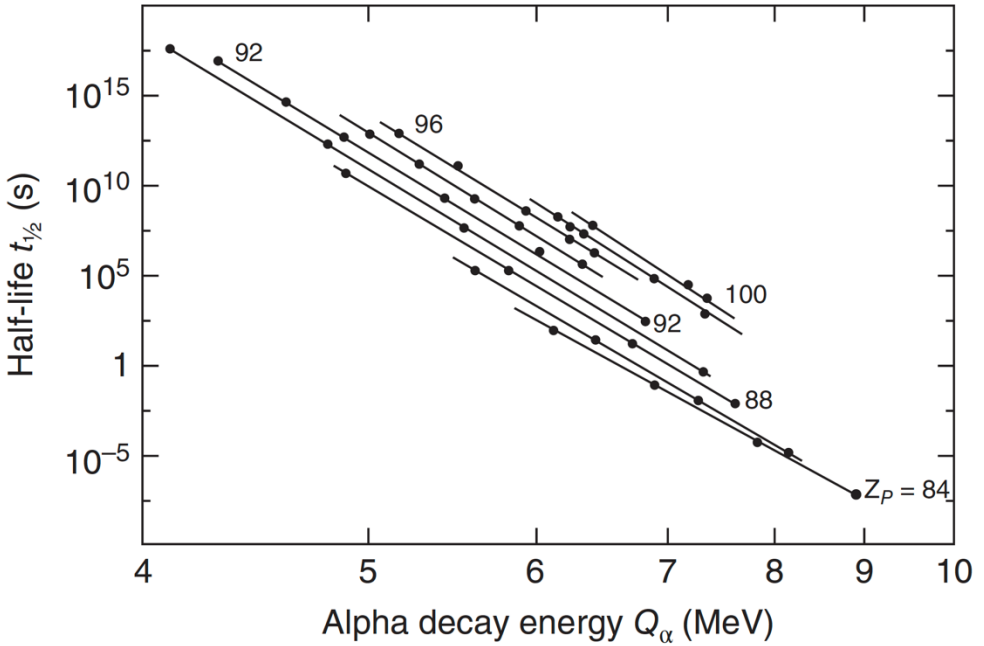
$$2G \approx 2\pi\frac{e^2}{\hbar c}Z_\alpha Z_{daughter}\sqrt{\frac{\mu c^2}{2Q_\alpha}}\left(1-\frac{4}{\pi}\sqrt{\frac{r_0\left(A_\alpha^{1/3}+A_{daughter}^{1/3}\right)}{\frac{Z_\alpha Z_{daughter}}{Q_\alpha}e^2}}\right)$$

- Since half-life is inversely proportional to the tunneling probability, $t_{1/2} \propto e^{2G} \propto e^{z/\sqrt{Q_\alpha}}$
- This is exactly what the Geiger-Nuttall relation already told us!

Qualitative Implications

- Q_α generally increases with increasing Z because there is increased penalty in the liquid drop model binding for increased Coulomb energy
- Increasing Z means increased Coulomb barrier height and therefore more of a barrier to tunnel through
  - For the same Q_α, increasing Z increases $t_{1/2}$
- Increasing Q_α means that the α energy required to tunnel through the barrier is larger, therefore the α bombards the barrier more frequently
  - For the same Z, Q_α decreases $t_{1/2}$

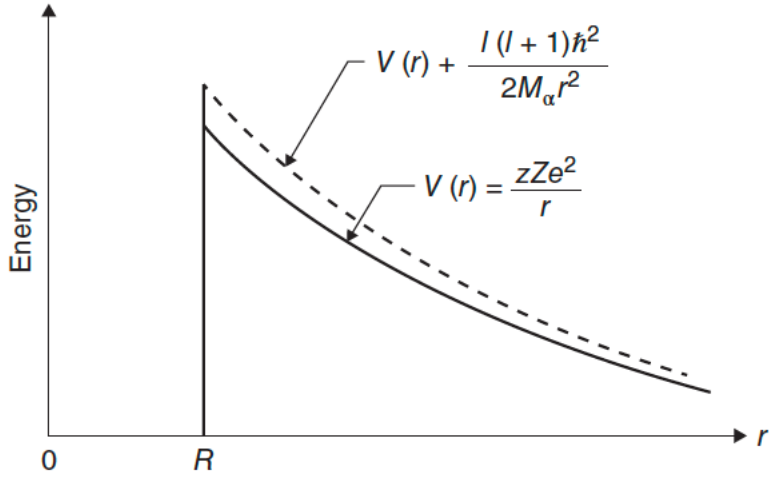




Impact of Centrifugal Barrier

- Alpha decay may involve transitioning from a nucleus with one J^{π} to another, meaning that the α particle carries away angular momentum
- This means that the α particle must tunnel through a centrifugal barrier:

$$V_{\ell} = \frac{\ell(\ell+1)\hbar^2}{2\mu R^2}$$

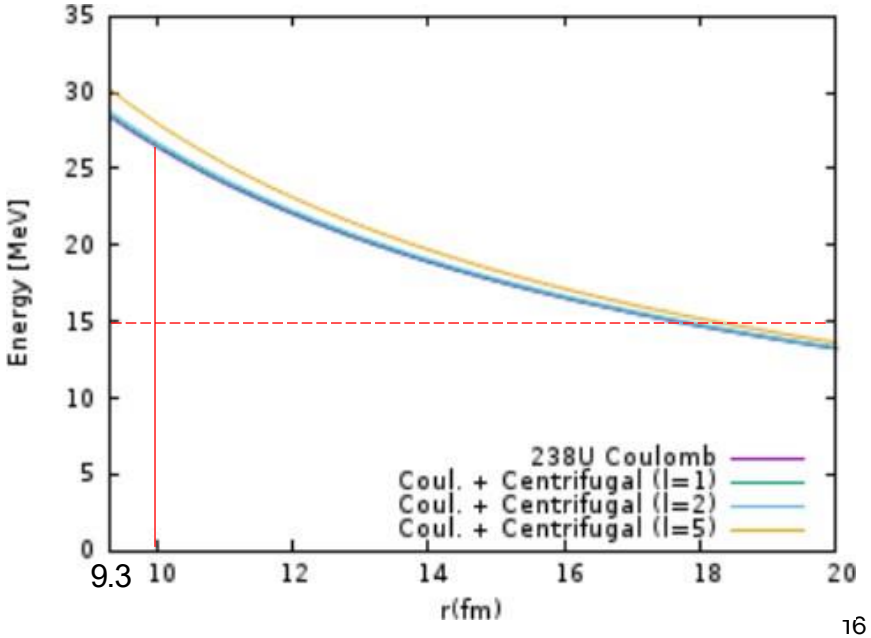


Where ℓ is the angular momentum carried away by the α particle

- Turns out that this correction is quite small

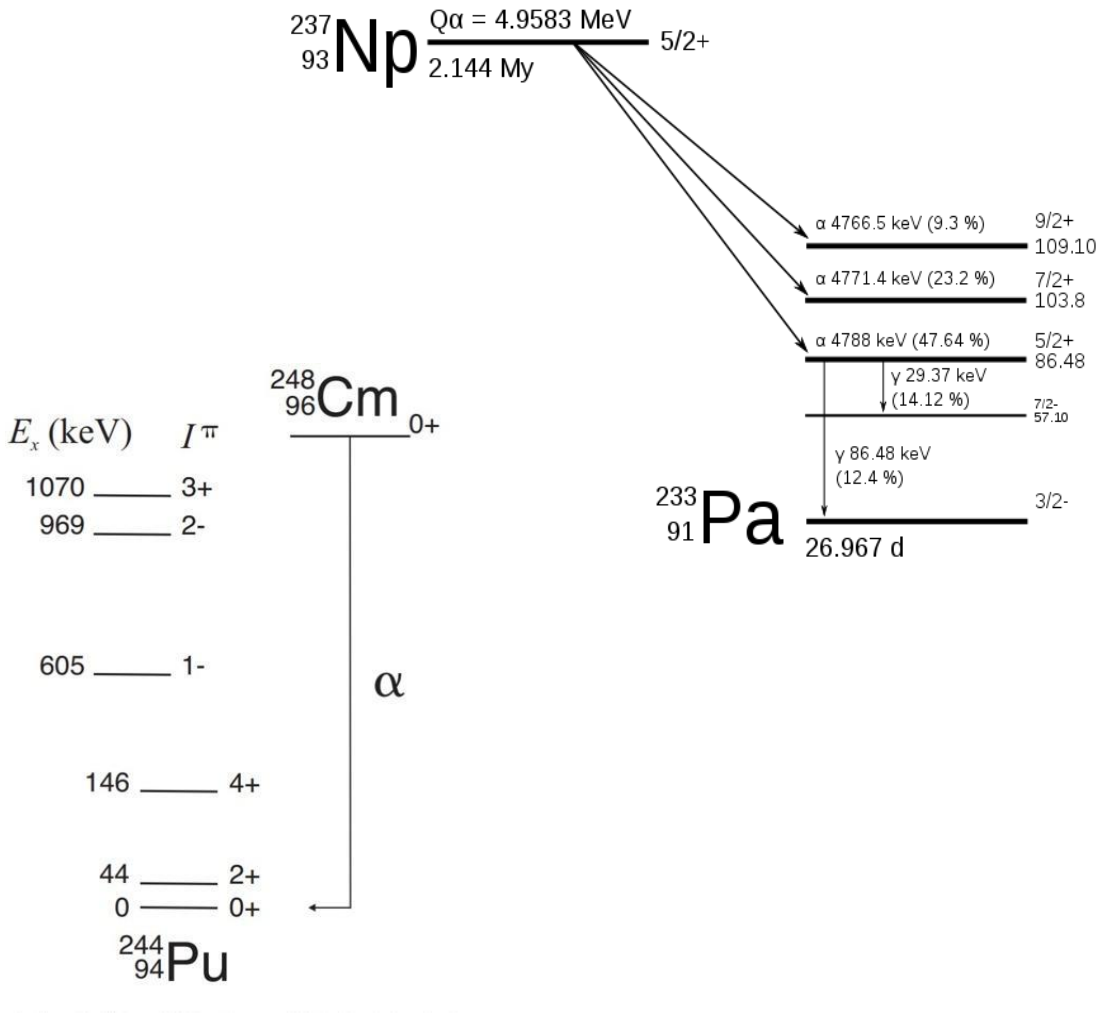
$$\lambda_{\ell\neq 0} \approx \lambda_{\ell=0} e^{(-2.027\ell(\ell+1)Z^{-1/2}A^{-1/6})}$$

- Compare this to the difference in barrier thickness corresponding to a fixed Q_{α} for a small change in R



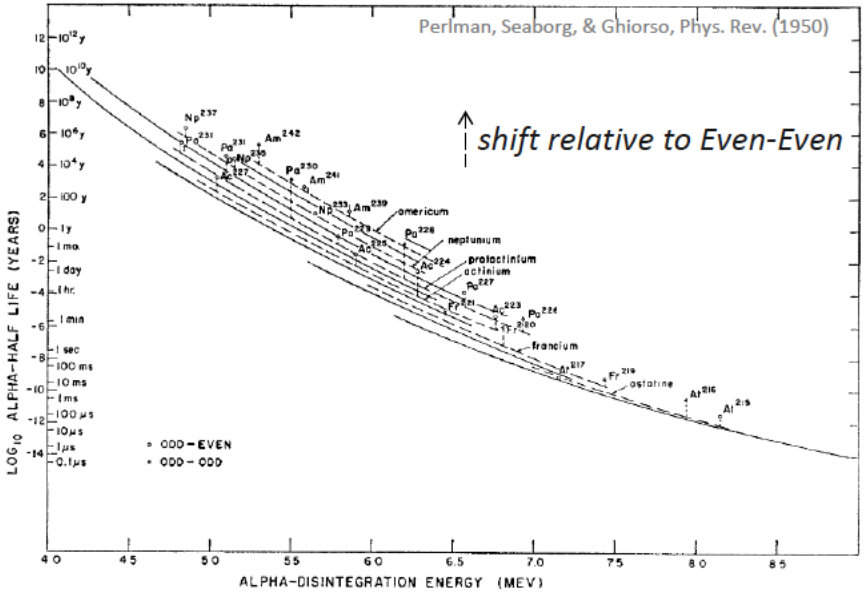
Impact of Parity Change on α Decay

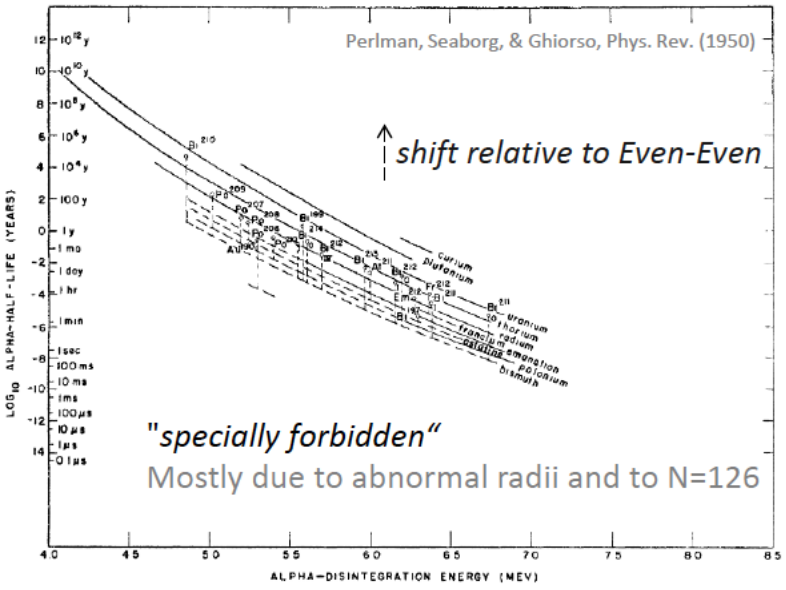
- Conservation of angular momentum and parity also places constraints on α decay processes
- Since the α particle has no intrinsic spin, the total angular momentum of the α particle must equal its orbital angular momentum ℓ with parity $\pi = (-1)^{\ell}$
- If the parent has $J^{\pi} = 0^{+}$ then allowed values of J^{π} in the daughter are $0^{+}\ (\ell = 0), 1^{-}\ (\ell = 1), 2^{+}\ (\ell = 2), \ldots$



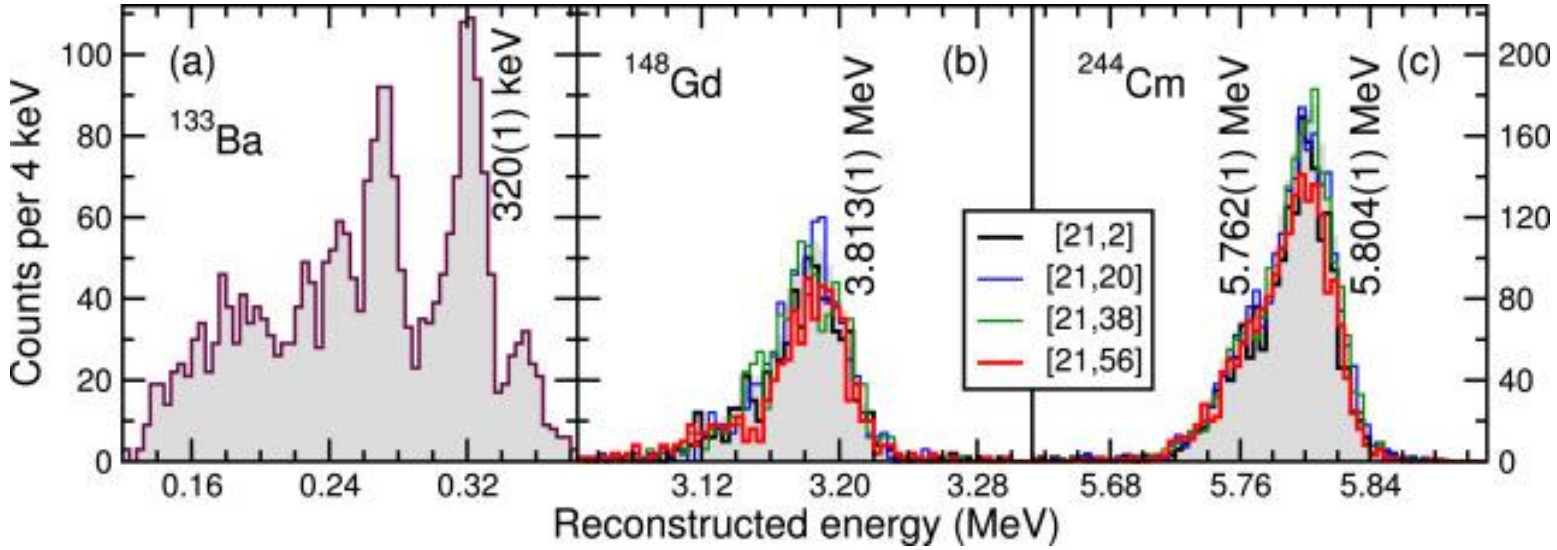
Hindrance factors

- Up to now, what we've discussed is relevant for even-Z, even-N nuclei
- For odd nuclei, the odd nucleon hinders α-decay by a factor of anywhere from a few to >1000
- For example, if the odd nucleon of the parent and daughter is in the same orbit, λ is reduced by a factor of ~4. If the parity changes, λ is reduced by a factor of ~100. If the spin and parity change, λ is reduced by a factor of ~1000



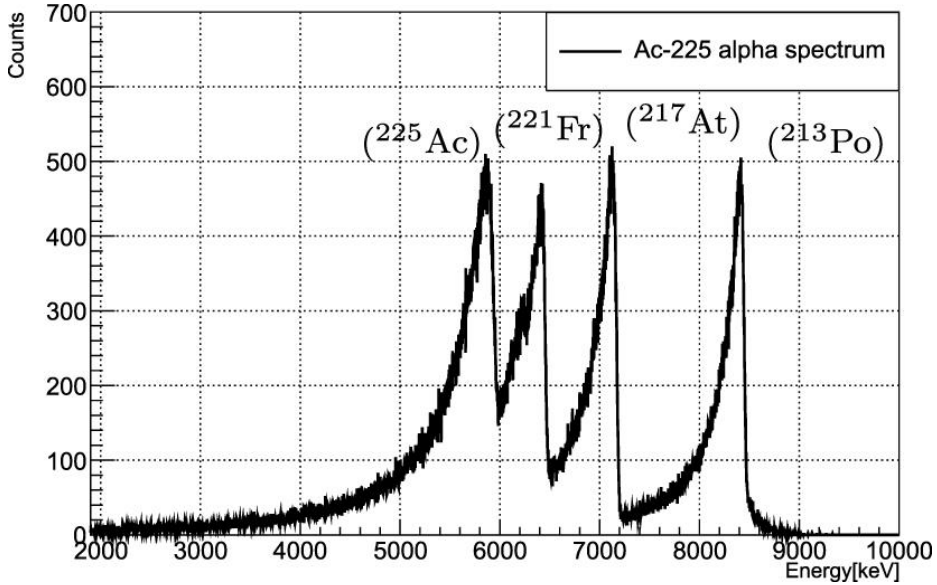


Alpha Spectroscopy



Spectroscopy of superheavy ^{244}Cm

P. Golubev *et al* *NIMA.* **1075** 170384 (2025)

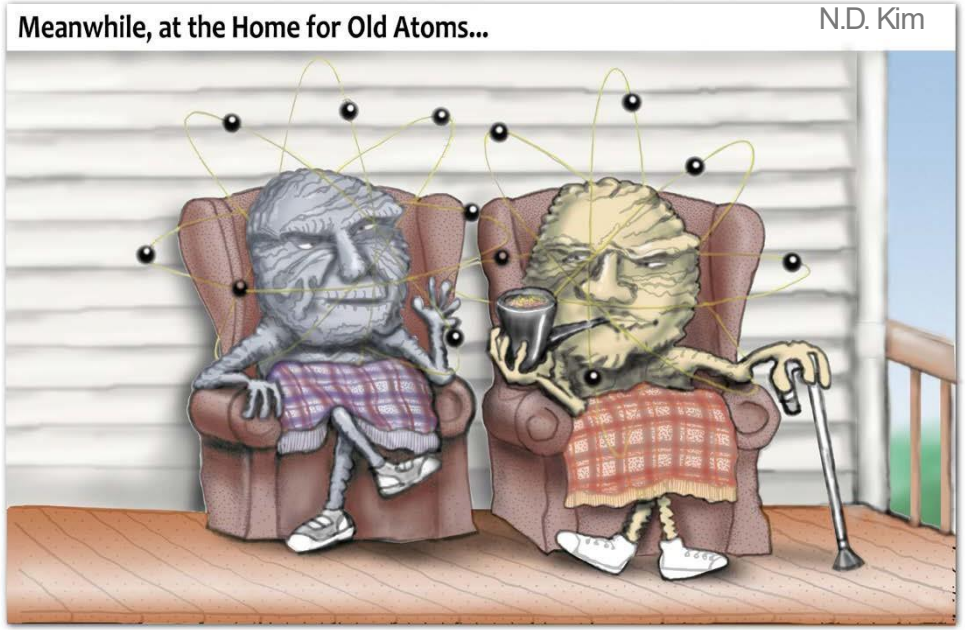


First spectroscopy of radiopharmaceutical ^{225}Ac

Mahsa Farasat *et al* *Phys. Med. Biol.* **70** 115019 (2025)

Why is it α particles that are being emitted?

- Why α decay? Why not proton decay, or ^{3}He decay, or ^{12}C decay?
- The short answer is Q-values, Coulomb barriers, and clustering probabilities
  - **Q-value**: The cluster decay must be energetically favorable
  - **Coulomb barrier**: Higher-Z particles will have a larger barrier to tunnel through
  - **Clustering probability:** It's less likely for more nucleons to congregate within a nucleus

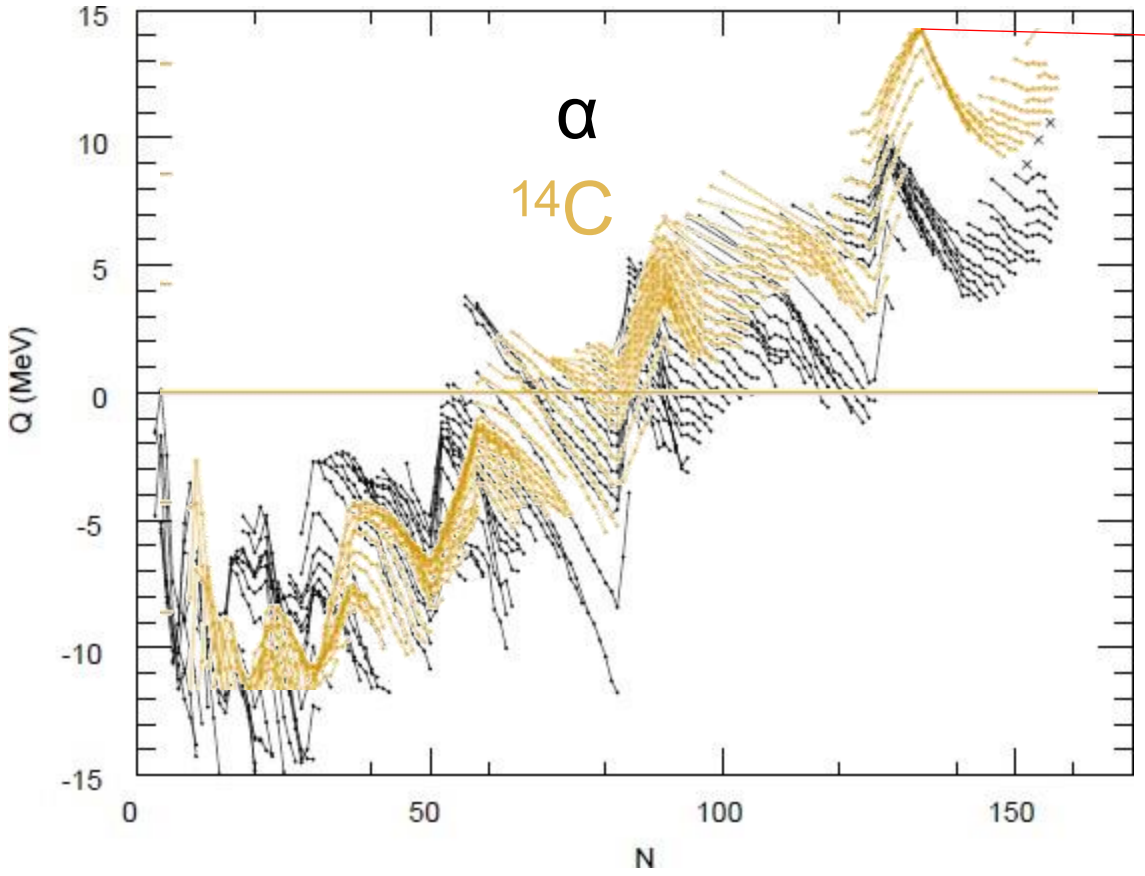


"When I was young I was an atom of uranium-238. I felt so alive, so dangerous! Then one day I accidentally ejected an alpha particle. Now look at me – a spent old atom of lead-206. All my life since then has been nothing but decay, decay, decay..."

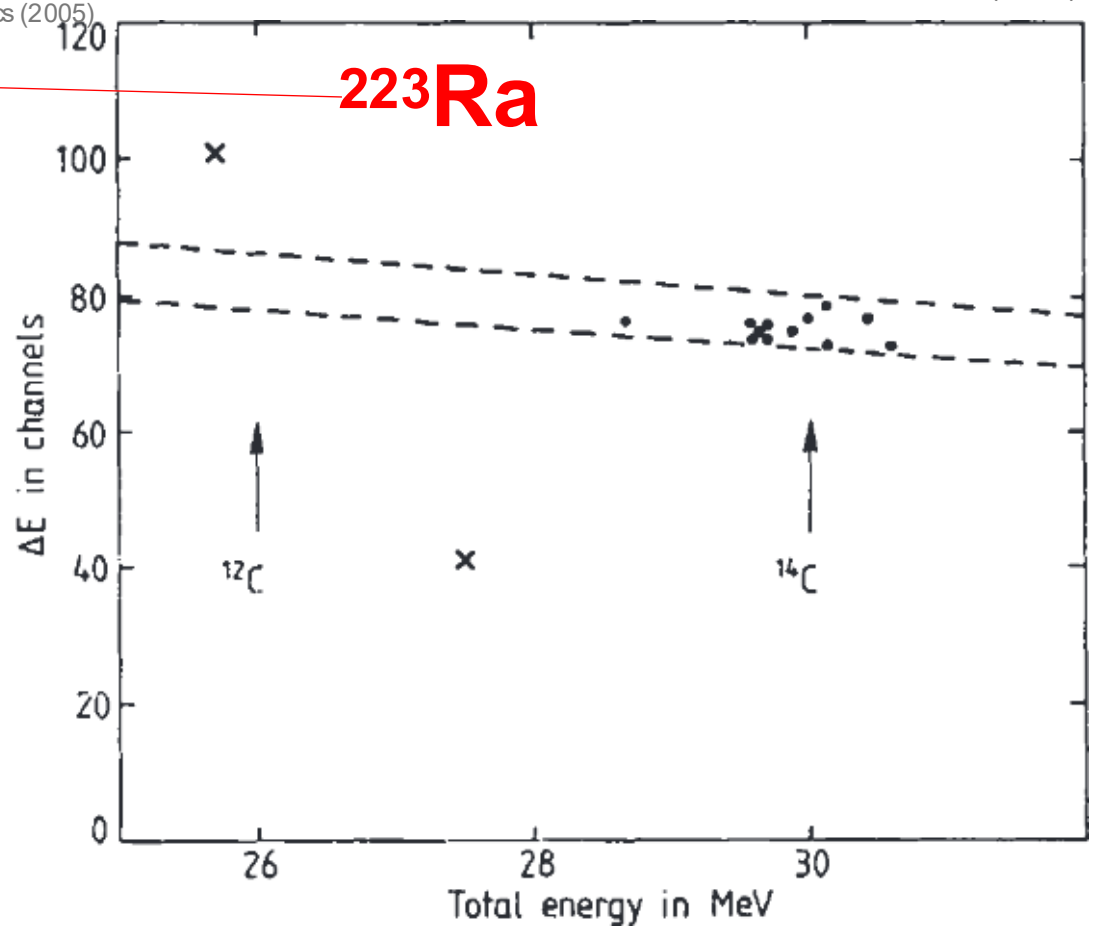
Why is it α particles that are being emitted? Q-value

- Decay is a spontaneous process that only occurs because there's a lower energy state that is available; i.e. a positive Q-value is required for a decay to occur
- The more positive the better (since this means more energy to tunnel)

B.A. Brown, Lecture Notes in Nuclear Structure Physics (2005)

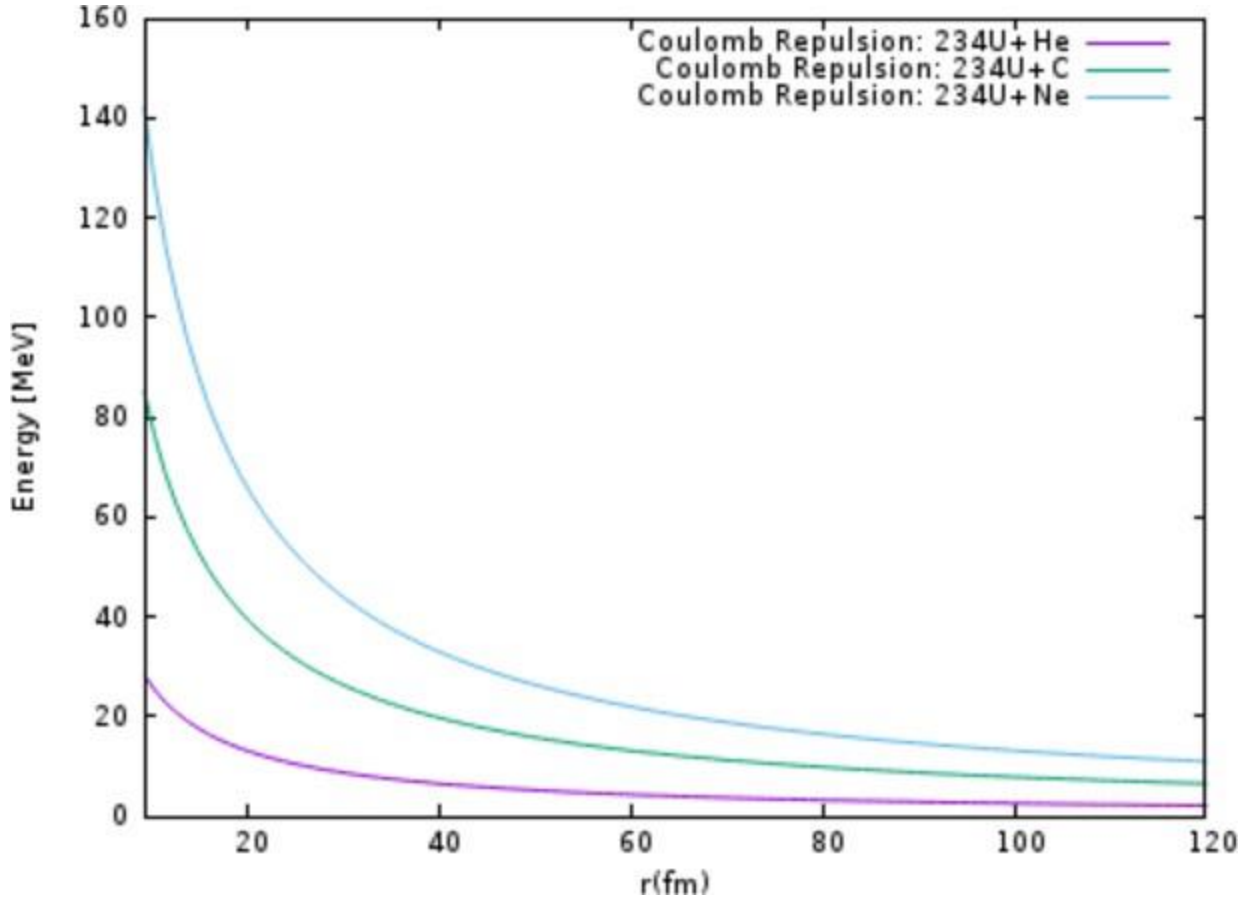


Rose & Jones, Nature (1984)



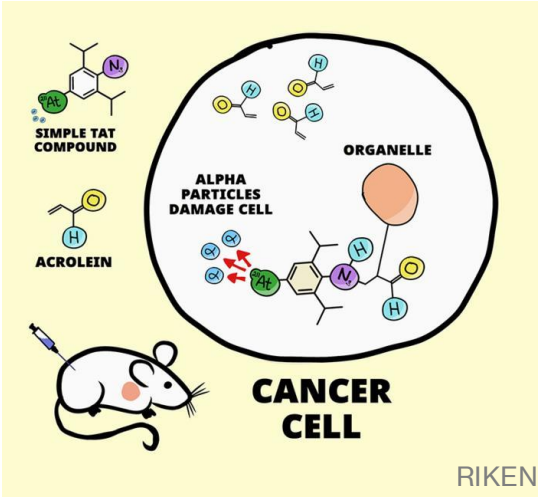
Why is it α particles that are being emitted? Coulomb

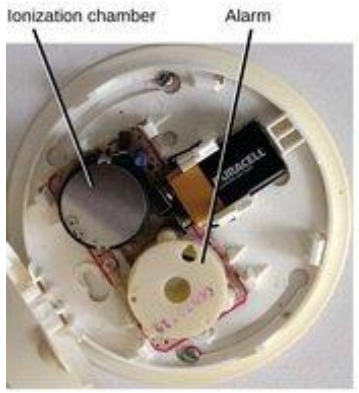
- The Coulomb barrier height scales with the charge of the particle being emitted
- It takes a much larger Q-value to make larger Z decay have any chance at tunneling through

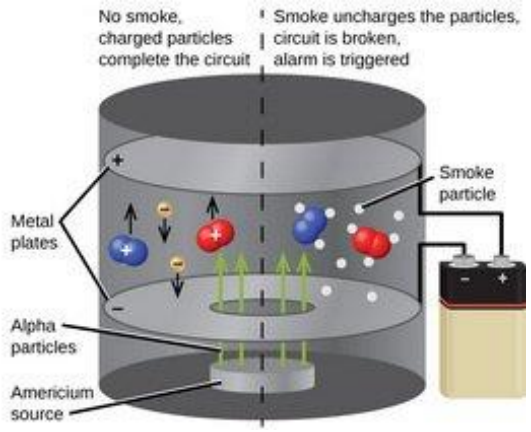


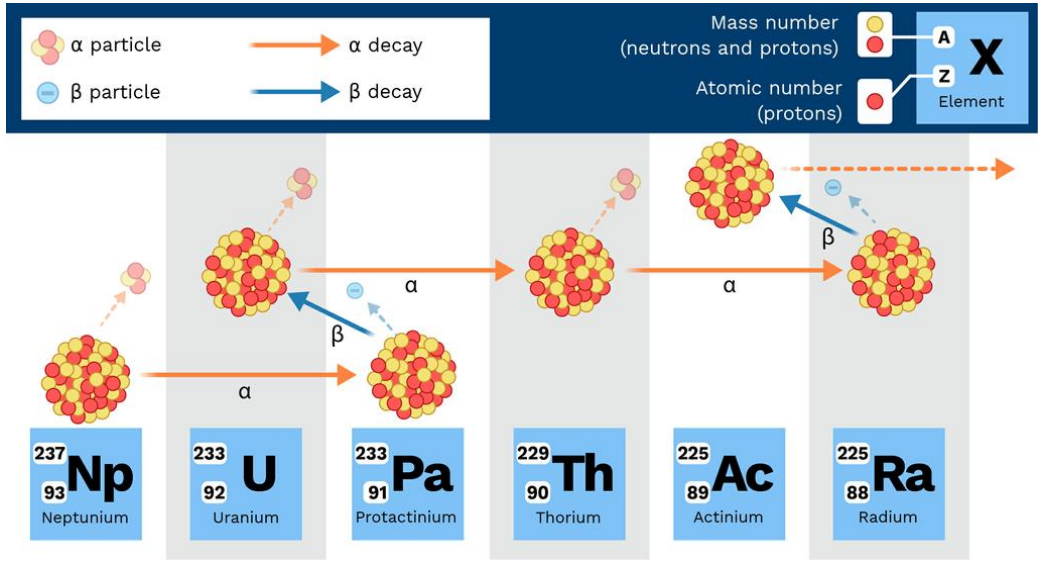
Why does α Decay Matter?

- It's a predominant decay mode for many heavy nuclei
- Nuclear Astrophysics (α-process)
- Applications
  - Smoke detectors contain ^{241}Am
  - Targeted Alpha Therapy to treat cancer
  - Decay chains are important to understand in Nuclear Reactors
  - Material science for measuring composition and thicknesses of samples









Wrapping up

- HW #2 due **Sept. 25** at 11:59PM
- Office Hours on Friday at 9:30-10:30AM
    - Edwards 204
- **Next class**: Beta Decay
- Questions?